SOFTWARE ENGINEERING LAB

Experiment No. 1

Preparation of Software Requirements Specification (SRS)
for Shield AI — Scam Detection System

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1. Aim

To prepare an SRS describing the functional and non-functional requirements of Shield AI, a system that analyses submitted text and flags potential scam messages.

2. Problem Statement

Scam messages delivered through SMS, email and other text channels often use urgency, impersonation, suspicious links or requests for sensitive information. Shield AI aims to help users assess such text by analysing its content and available URL signals, then presenting a risk verdict with understandable reasons. The system supports awareness and does not guarantee that every scam will be detected.

3. Software Requirements Specification (SRS)

3.1 Purpose

Describe the purpose of the proposed Shield AI system.

Shield AI helps users identify potentially fraudulent or suspicious messages by analysing text pasted from email, SMS, chat or other sources. It returns a risk indication and reasons, helping users make a more informed decision before clicking links or sharing information.

3.2 Scope

Describe the major functions included within the scope of the system.

The system accepts user-provided text, preprocesses it, checks for scam-related language patterns and suspicious URLs, combines available signals into a risk assessment, and displays a verdict with an explanation. It may keep scan history and accept user feedback where those features are enabled. It does not automatically access private inboxes unless a separate integration is implemented and authorised.

3.3 Stakeholders / User Classes

S. No.	Stakeholder / User	Role
1	End User	Submits text and reviews risk assessment
2	System Administrator	Maintains detection rules and system configuration
3	Threat Intelligence / URL Reputation Service	Provides URL/domain reputation data when available
4	Project Developer / Maintainer	Maintains application, detection logic and updates

3.4 Assumptions

1. The user provides the text to analyse and has permission to share it with Shield AI.
2. URL reputation checks depend on availability of an external service or locally configured rules.
3. Detection results are indicative and may include false positives or false negatives.
4. The initial version analyses submitted text; automatic mailbox/SMS access is outside scope unless explicitly added.

4. Functional Requirements

Identify at least 8 functional requirements. Write each as a clear statement using “The system shall …”.

ID	Functional Requirement	Priority
FR-01	The system shall accept text copied from email, SMS, chat or another text source for analysis.	High
FR-02	The system shall validate that the submitted input is not empty and is within the configured size limit.	High
FR-03	The system shall preprocess submitted text for analysis while preserving its relevant meaning.	High
FR-04	The system shall identify suspicious phrases, urgency, impersonation cues and requests for sensitive information.	High
FR-05	The system shall extract URLs or domains from submitted text when present.	High
FR-06	The system shall use configured rules or an available reputation service to assess suspicious URLs/domains.	Medium
FR-07	The system shall display a risk verdict or category with a score/confidence indicator where supported and an explanation of key signals.	High
FR-08	The system shall allow the user to view recent scan history and submit feedback if those features are enabled.	Medium

5. Non-Functional Requirements

Identify at least 4 non-functional requirements related to performance, security, usability, reliability, etc.

ID	Category	Non-Functional Requirement
NFR-01	Performance	For normal-sized text, the system should return an assessment within 5 seconds under typical conditions, excluding delays from external services.
NFR-02	Security	The system shall avoid exposing submitted text to unauthorised users and shall not request passwords, OTPs or banking credentials for analysis.
NFR-03	Usability	The interface shall provide a clear input area, an obvious scan action and an understandable result explanation.
NFR-04	Reliability / Backup	The system shall handle unavailable reputation services gracefully and shall not report an external lookup as successful when it fails.

6. Constraints / Business Rules

Identify at least 4 constraints or business rules.

ID	Constraint / Business Rule
BR-01	Shield AI analyses only text intentionally submitted by the user in the initial scope.
BR-02	A high-risk verdict is a warning, not definitive proof of fraud; users should verify through trusted official channels.
BR-03	External URL reputation is used only when configured and available; otherwise the system must indicate that the lookup was unavailable or use local signals.
BR-04	Passwords, OTPs, full payment-card details and other secrets should not be required as input for scam detection.

7. External Interface Requirements

7.1 User Interface

Mention the major screens/interfaces required in the system (e.g., Login, Search Book, Issue/Return).

The main interface includes a text submission screen, Scan button, result view showing verdict/risk indicators and reasons, optional scan history, and optional feedback form. Administrative configuration is limited to authorised maintainers if included.

7.2 Input Requirements

S. No.	Input	Description
1	Message text	Text copied from SMS, email, chat or other message
2	URLs / domains	Extracted from submitted text when present
3	User feedback	Optional label such as “likely scam” or “not a scam” for a prior scan
4	Configuration	Detection rules and reputation-service settings, managed by authorised maintainers

7.3 Output Requirements

S. No.	Output	Description
1	Risk verdict	Potential scam / suspicious / lower-risk indication based on configured logic
2	Risk explanation	Signals such as urgent wording, credential requests or suspicious URLs
3	URL assessment	Available reputation result or a clear indication that lookup was unavailable
4	Scan history / confirmation	Optional prior scan summary or feedback-submission confirmation

8. Requirement Review

1. Are all requirements clear and understandable?

Yes. Requirements use specific actions and observable outputs, such as accepting text and displaying a risk verdict.

2. Are any requirements contradictory?

No direct contradictions are identified. Requirements distinguish optional features and external-service availability from core text analysis.

3. Are the requirements testable? Give one example.

Yes. Example: submit a non-empty text containing a suspicious URL and verify that the result includes an assessment and the URL-related signal, or an explicit lookup-unavailable message.

9. Result

Write 2–3 lines describing what was accomplished in this experiment.

The SRS for Shield AI was prepared with its scope, stakeholders, assumptions, functional and non-functional requirements, business rules, and input/output interfaces. These requirements provide a basis for design and testing.